Seyed Alireza Mousavian Anaraki

Nationality: Iranian

Date of birth: 22/11/1996

Sheffield, UK

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Google Scholar

Research Gate

LinkedIn

EDUCATION

- 01/11/2023-Present

  PhD student in Data Science at university of Rome Tor Vergata, Rome, Italy
  Research Focus: Natural Language Processing (NLP) for sustainability (ESG) analysis, particularly on large-scale processing of sustainability reports, weak-supervision techniques, and the use of large language models (LLM) for structure, consistency, and credibility assessment.
  Supervisors: Profs. Danilo Croce and Roberto Basili **Email:** {croce, basili}@dii.uniroma2.it

- 23/09/2019 – 13/11/2021

  Master's degree in Industrial Engineering (Engineering Management), Iran University of Science and Technology, Department of Industrial Engineering, Tehran, Iran
  legal duration of the course= 2 years, GPA: 19.33/20 (**Ranked 1st** student in Master GPA)
  Thesis Title: Development and implementation of data mining methods by creating strong, balanced, and interpretable clusters to adopt appropriate policies
  Supervisors: Dr. Abdorrahman Haeri **Email:** ahaeri@iust.ac.ir Google Scholar

- 23/09/2015 – 27/08/2019

  Bachelor's degree in Industrial Engineering, Kashan University, Department of Industrial Engineering, Kashan, Iran
  legal duration of the course= 4 years, GPA: 18.62/20 (**Ranked 1st** student in Bachelor GPA)

RESEARCH INTERESTS

- ✓ Artificial Intelligence (AI), Deep Learning (DL), and Machine Learning (ML)
- ✓ AI-driven decision-making in management
- ✓ NLP and LLMs for sustainability (ESG) reporting
- ✓ Operations research applications

TEACHING ASSISTANT

- 23/09/2017 – 19/02/2018

  Teacher assistant for **Linear Algebra** course at Kashan university, Kashan, Iran
  Supervisor: Dr. Mohammad Taghi Rezvan
  Email: rezvan@kashanu.ac.ir

RESEARCH ASSISTANT

- 21/11/2020 – 22/08/2021

Researcher and data analyzer at Iran's National Elites Foundation (INEF), Tehran, Iran.

- ✓ Participated in data analytics research group as **data-driven reliability-centered maintenance specialist** for critical **energy infrastructure maintenance improvement**.

Supervisor: Dr. Alireza Fereidunian **Email:** fereidunian@kntu.ac.ir

- 23/09/2019 – 13/11/2021

Research assistant at Iran University of Science and Technology,Tehran, Iran

- ✓ Participated as a research assistant on a team led by Dr. Abdorrahman Haeri to **develop and implement data mining methods in different fields with an emphasis on data-driven decision-making**.

Supervisors: Dr. Abdorrahman Haeri **Email:** ahaeri@iust.ac.ir

RESEARCH ACHIEVEMENTS

Journal Papers

- ✓ Anaraki, S. A. M., Croce, D., & Basili, R. (2025). Large language models for sustainability reporting: A systematic review and research agenda. *Sustainable Futures*, *10*, 101494. https://doi.org/10.1016/j.sftr.2025.101494
- ✓ Fallahi, A., Mousavian Anaraki, S. A., Mokhtari, H., & Niaki, S. T. A. (2022). Blood plasma supply chain planning to respond COVID-19 pandemic: a case study. *Environment, Development and Sustainability*, 1-52. https://doi.org/10.1007/s10668-022-02793-7
- ✓ Anaraki, S. A. M., & Haeri, A. (2022). Soft and hard hybrid balanced clustering with innovative qualitative balancing approach. *Information Sciences*. https://doi.org/10.1016/j.ins.2022.09.044
- ✓ Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2022). Generating balanced and strong clusters based on balance-constrained clustering approach (strong balance-constrained clustering) for improving ensemble classifier performance. *Neural Computing and Applications*, 1-17. https://doi.org/10.1007/s00521-022-07595-6
- ✓ Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2021). A hybrid reciprocal model of PCA and K-means with an innovative approach of considering sub-datasets for the improvement of K-means initialization and step-by-step labeling to create clusters with high interpretability. *Pattern Analysis and Applications*, 24(3), 1387–1402. https://doi.org/10.1007/s10044-021-00977-x
- ✓ Mousavian, S. A., Haeri, A., & Moslehi, F. (2021). Providing a Hybrid Clustering Method as an Auxiliary System in Automatic Labeling to Divide Employee into Different Levels of Productivity and their Retention. Iranian Journal of Management Studies. https://doi:10.22059/ijms.2021.299705.674004
- ✓ Rahimi, F., & Mousavian Anaraki, S. A. (2020). Proposing an Innovative Model Based on the Sierpinski Triangle for Forecasting EUR/USD Direction Changes. Journal of Money and Economy, 15(4), 423–444. https://doi.org/10.52547/jme.15.4.423

Conference Papers

- ✓ Anaraki, S. A. M., Croce, D., & Basili, R. (2025, September). Automatic GRI-SDG Annotation and LLM-Based Filtering for Sustainability Reports. In *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)* (pp. 775-784).

- ✓ Anaraki, S. A. M., Croce, D., & Basili, R. (2025, July). Unsupervised Sustainability Report Labeling based on the integration of the GRI and SDG standards. *In Proceedings of the Fourth Workshop on NLP for Positive Impact (NLP4PI)* (pp. 151-162). https://aclanthology.org/2025.nlp4pi-1.13/

HONORS AND AWARDS

- July 2025

  Outstanding Paper Award, Workshop on NLP for Positive Impact (NLP4PI @ ACL 2025)

- July 2020

  Ranked 1st student in Master GPA among near 20 students in the "Department of Industrial Engineering" of Iran University of Science and Technology

- September 2019

  Eligible to continue my education in the Master's Program at the Iran University of Science and Technology without taking the "National University Entrance Exam"

- July 2018

  Ranked 1st student in Bachelor GPA among near 40 students in the "Department of Industrial Engineering" of Kashan university

LANGUAGE SKILLS

- **English:** Upper Intermediate
- **Persian:** Native

COMPUTER SKILLS

- **Programming Software:**

  Python, R

  Database management system: SQL Server

- **Engineering Software:**

  Data visualization/BI tools: Tableau, Power BI
  Other: Gams, COMFAR, Expert Choice, ALDEP, VOS viewer